

Gameboy CPU (LR35902) instruction set

	x0	x1	x2	x3	x4	x5	x6	x7	x8	x9	xA	xB	xC	xD	xE	xF
0x	NOP 1 4 -- --	LD BC,d16 3 12 -- --	LD (BC),A 1 8 -- --	INC BC 1 8 -- --	INC B 1 4 Z 0 H -	DEC B 1 4 Z 1 H -	LD B,d8 2 8 -- --	RLCA 1 4 0 0 0 C	LD (a16),SP 3 20 -- --	ADD HL,BC 1 8 -- 0 H C	LD A,(BC) 1 8 -- --	DEC BC 1 8 -- --	INC C 1 4 Z 0 H -	DEC C 1 4 Z 1 H -	LD C,d8 2 8 -- --	RRCA 1 4 0 0 0 C
1x	STOP 0 2 4 -- --	LD DE,d16 3 12 -- --	LD (DE),A 1 8 -- --	INC DE 1 8 -- --	INC D 1 4 Z 0 H -	DEC D 1 4 Z 1 H -	LD D,d8 2 8 -- --	RLA 1 4 0 0 0 C	JR r8 2 12 -- --	ADD HL,DE 1 8 -- 0 H C	LD A,(DE) 1 8 -- --	DEC DE 1 8 -- --	INC E 1 4 Z 0 H -	DEC E 1 4 Z 1 H -	LD E,d8 2 8 -- --	RRA 1 4 0 0 0 C
2x	JR NZ,r8 2 12/8 -- --	LD HL,d16 3 12 -- --	LD (HL+),A 1 8 -- --	INC HL 1 8 -- --	INC H 1 4 Z 0 H -	DEC H 1 4 Z 1 H -	LD H,d8 2 8 -- --	DAA 1 4 Z - 0 C	JR Z,r8 2 12/8 -- --	ADD HL,HL 1 8 -- 0 H C	LD A,(HL+) 1 8 -- --	DEC HL 1 8 -- --	INC L 1 4 Z 0 H -	DEC L 1 4 Z 1 H -	LD L,d8 2 8 -- --	CPL 1 4 - 1 1 -
3x	JR NC,r8 2 12/8 -- --	LD SP,d16 3 12 -- --	LD (HL-),A 1 8 -- --	INC SP 1 8 -- --	INC (HL) 1 12 Z 0 H -	DEC (HL) 1 12 Z 1 H -	LD (HL),d8 2 12 -- --	SCF 1 4 - 0 0 1	JR C,r8 2 12/8 -- --	ADD HL,SP 1 8 -- 0 H C	LD A,(HL-) 1 8 -- --	DEC SP 1 8 -- --	INC A 1 4 Z 0 H -	DEC A 1 4 Z 1 H -	LD A,d8 2 8 -- --	CCF 1 4 - 0 0 C
4x	LD B,B 1 4 -- --	LD B,C 1 4 -- --	LD B,D 1 4 -- --	LD B,E 1 4 -- --	LD B,H 1 4 -- --	LD B,L 1 4 -- --	LD B,(HL) 1 8 -- --	LD B,A 1 4 -- --	LD C,B 1 4 -- --	LD C,C 1 4 -- --	LD C,D 1 4 -- --	LD C,E 1 4 -- --	LD C,H 1 4 -- --	LD C,L 1 4 -- --	LD C,(HL) 1 8 -- --	LD C,A 1 4 -- --
5x	LD D,B 1 4 -- --	LD D,C 1 4 -- --	LD D,D 1 4 -- --	LD D,E 1 4 -- --	LD D,H 1 4 -- --	LD D,L 1 4 -- --	LD D,(HL) 1 8 -- --	LD D,A 1 4 -- --	LD E,B 1 4 -- --	LD E,C 1 4 -- --	LD E,D 1 4 -- --	LD E,E 1 4 -- --	LD E,H 1 4 -- --	LD E,L 1 4 -- --	LD E,(HL) 1 8 -- --	LD E,A 1 4 -- --
6x	LD H,B 1 4 -- --	LD H,C 1 4 -- --	LD H,D 1 4 -- --	LD H,E 1 4 -- --	LD H,H 1 4 -- --	LD H,L 1 4 -- --	LD H,(HL) 1 8 -- --	LD H,A 1 4 -- --	LD L,B 1 4 -- --	LD L,C 1 4 -- --	LD L,D 1 4 -- --	LD L,E 1 4 -- --	LD L,H 1 4 -- --	LD L,L 1 4 -- --	LD L,(HL) 1 8 -- --	LD L,A 1 4 -- --
7x	LD (HL),B 1 8 -- --	LD (HL),C 1 8 -- --	LD (HL),D 1 8 -- --	LD (HL),E 1 8 -- --	LD (HL),H 1 8 -- --	LD (HL),L 1 8 -- --	HALT 1 4 -- --	LD (HL),A 1 8 -- --	LD A,B 1 4 -- --	LD A,C 1 4 -- --	LD A,D 1 4 -- --	LD A,E 1 4 -- --	LD A,H 1 4 -- --	LD A,L 1 4 -- --	LD A,(HL) 1 8 -- --	LD A,A 1 4 -- --
8x	ADD A,B 1 4 Z 0 H C	ADD A,C 1 4 Z 0 H C	ADD A,D 1 4 Z 0 H C	ADD A,E 1 4 Z 0 H C	ADD A,H 1 4 Z 0 H C	ADD A,L 1 4 Z 0 H C	ADD A,(HL) 1 8 Z 0 H C	ADD A,A 1 4 Z 0 H C	ADC A,B 1 4 Z 0 H C	ADC A,C 1 4 Z 0 H C	ADC A,D 1 4 Z 0 H C	ADC A,E 1 4 Z 0 H C	ADC A,H 1 4 Z 0 H C	ADC A,L 1 4 Z 0 H C	ADC A,(HL) 1 8 Z 0 H C	ADC A,A 1 4 Z 0 H C
9x	SUB B 1 4 Z 1 H C	SUB C 1 4 Z 1 H C	SUB D 1 4 Z 1 H C	SUB E 1 4 Z 1 H C	SUB H 1 4 Z 1 H C	SUB L 1 4 Z 1 H C	SUB (HL) 1 8 Z 1 H C	SUB A 1 4 Z 1 H C	SBC A,B 1 4 Z 1 H C	SBC A,C 1 4 Z 1 H C	SBC A,D 1 4 Z 1 H C	SBC A,E 1 4 Z 1 H C	SBC A,H 1 4 Z 1 H C	SBC A,L 1 4 Z 1 H C	SBC A,(HL) 1 8 Z 1 H C	SBC A,A 1 4 Z 1 H C
Ax	AND B 1 4 Z 0 1 0	AND C 1 4 Z 0 1 0	AND D 1 4 Z 0 1 0	AND E 1 4 Z 0 1 0	AND H 1 4 Z 0 1 0	AND L 1 4 Z 0 1 0	AND (HL) 1 8 Z 0 1 0	AND A 1 4 Z 0 1 0	XOR B 1 4 Z 0 0 0	XOR C 1 4 Z 0 0 0	XOR D 1 4 Z 0 0 0	XOR E 1 4 Z 0 0 0	XOR H 1 4 Z 0 0 0	XOR L 1 4 Z 0 0 0	XOR (HL) 1 8 Z 0 0 0	XOR A 1 4 Z 0 0 0
Bx	OR B 1 4 Z 0 0 0	OR C 1 4 Z 0 0 0	OR D 1 4 Z 0 0 0	OR E 1 4 Z 0 0 0	OR H 1 4 Z 0 0 0	OR L 1 4 Z 0 0 0	OR (HL) 1 8 Z 0 0 0	OR A 1 4 Z 0 0 0	CP B 1 4 Z 1 H C	CP C 1 4 Z 1 H C	CP D 1 4 Z 1 H C	CP E 1 4 Z 1 H C	CP H 1 4 Z 1 H C	CP L 1 4 Z 1 H C	CP (HL) 1 8 Z 1 H C	CP A 1 4 Z 1 H C
Cx	RET NZ 1 20/8 -- --	POP BC 1 12 -- --	JP NZ,a16 3 16/12 -- --	JP a16 3 16 -- --	CALL NZ,a16 3 24/12 -- --	PUSH BC 1 16 -- --	ADD A,d8 2 8 Z 0 H C	RST 00H 1 16 -- --	RET Z 1 20/8 -- --	RET 1 16 -- --	JP Z,a16 3 16/12 -- --	PREFIX CB 1 4 -- --	CALL Z,a16 3 24/12 -- --	CALL a16 3 24 -- --	ADC A,d8 2 8 Z 0 H C	RST 08H 1 16 -- --
Dx	RET NC 1 20/8 -- --	POP DE 1 12 -- --	JP NC,a16 3 16/12 -- --		CALL NC,a16 3 24/12 -- --	PUSH DE 1 16 -- --	SUB d8 2 8 Z 1 H C	RST 10H 1 16 -- --	RET C 1 20/8 -- --	RETI 1 16 -- --	JP C,a16 3 16/12 -- --		CALL C,a16 3 24/12 -- --		SBC A,d8 2 8 Z 1 H C	RST 18H 1 16 -- --
Ex	LDH (a8),A 2 12 -- --	POP HL 1 12 -- --	LD (C),A 2 8 -- --			PUSH HL 1 16 -- --	AND d8 2 8 Z 0 1 0	RST 20H 1 16 -- --	ADD SP,r8 2 16 0 0 H C	JP (HL) 1 4 -- --	LD (a16),A 3 16 -- --				XOR d8 2 8 Z 0 0 0	RST 28H 1 16 -- --
Fx	LDH A,(a8) 2 12 -- --	POP AF 1 12 Z N H C	LD A,(C) 2 8 -- --	DI 1 4 -- --		PUSH AF 1 16 -- --	OR d8 2 8 Z 0 0 0	RST 30H 1 16 -- --	LD HL,SP+r8 2 12 0 0 H C	LD SP,HL 1 8 -- --	LD A,(a16) 3 16 -- --	EI 1 4 -- --			CP d8 2 8 Z 1 H C	RST 38H 1 16 -- --

Prefix CB

	x0	x1	x2	x3	x4	x5	x6	x7	x8	x9	xA	xB	xC	xD	xE	xF
0x	RLC B 2 8 Z 0 0 C	RLC C 2 8 Z 0 0 C	RLC D 2 8 Z 0 0 C	RLC E 2 8 Z 0 0 C	RLC H 2 8 Z 0 0 C	RLC L 2 8 Z 0 0 C	RLC (HL) 2 16 Z 0 0 C	RLC A 2 8 Z 0 0 C	RRC B 2 8 Z 0 0 C	RRC C 2 8 Z 0 0 C	RRC D 2 8 Z 0 0 C	RRC E 2 8 Z 0 0 C	RRC H 2 8 Z 0 0 C	RRC L 2 8 Z 0 0 C	RRC (HL) 2 16 Z 0 0 C	RRC A 2 8 Z 0 0 C
1x	RL B 2 8 Z 0 0 C	RL C 2 8 Z 0 0 C	RL D 2 8 Z 0 0 C	RL E 2 8 Z 0 0 C	RL H 2 8 Z 0 0 C	RL L 2 8 Z 0 0 C	RL (HL) 2 16 Z 0 0 C	RL A 2 8 Z 0 0 C	RR B 2 8 Z 0 0 C	RR C 2 8 Z 0 0 C	RR D 2 8 Z 0 0 C	RR E 2 8 Z 0 0 C	RR H 2 8 Z 0 0 C	RR L 2 8 Z 0 0 C	RR (HL) 2 16 Z 0 0 C	RR A 2 8 Z 0 0 C
2x	SLA B 2 8 Z 0 0 C	SLA C 2 8 Z 0 0 C	SLA D 2 8 Z 0 0 C	SLA E 2 8 Z 0 0 C	SLA H 2 8 Z 0 0 C	SLA L 2 8 Z 0 0 C	SLA (HL) 2 16 Z 0 0 C	SLA A 2 8 Z 0 0 C	SRA B 2 8 Z 0 0 0	SRA C 2 8 Z 0 0 0	SRA D 2 8 Z 0 0 0	SRA E 2 8 Z 0 0 0	SRA H 2 8 Z 0 0 0	SRA L 2 8 Z 0 0 0	SRA (HL) 2 16 Z 0 0 0	SRA A 2 8 Z 0 0 0

3x	SWAP B 2 8 Z 0 0 0	SWAP C 2 8 Z 0 0 0	SWAP D 2 8 Z 0 0 0	SWAP E 2 8 Z 0 0 0	SWAP H 2 8 Z 0 0 0	SWAP L 2 8 Z 0 0 0	SWAP (HL) 2 16 Z 0 0 0	SWAP A 2 8 Z 0 0 0	SRL B 2 8 Z 0 0 C	SRL C 2 8 Z 0 0 C	SRL D 2 8 Z 0 0 C	SRL E 2 8 Z 0 0 C	SRL H 2 8 Z 0 0 C	SRL L 2 8 Z 0 0 C	SRL (HL) 2 16 Z 0 0 C	SRL A 2 8 Z 0 0 C
4x	BIT 0,B 2 8 Z 0 1 -	BIT 0,C 2 8 Z 0 1 -	BIT 0,D 2 8 Z 0 1 -	BIT 0,E 2 8 Z 0 1 -	BIT 0,H 2 8 Z 0 1 -	BIT 0,L 2 8 Z 0 1 -	BIT 0,(HL) 2 16 Z 0 1 -	BIT 0,A 2 8 Z 0 1 -	BIT 1,B 2 8 Z 0 1 -	BIT 1,C 2 8 Z 0 1 -	BIT 1,D 2 8 Z 0 1 -	BIT 1,E 2 8 Z 0 1 -	BIT 1,H 2 8 Z 0 1 -	BIT 1,L 2 8 Z 0 1 -	BIT 1,(HL) 2 16 Z 0 1 -	BIT 1,A 2 8 Z 0 1 -
5x	BIT 2,B 2 8 Z 0 1 -	BIT 2,C 2 8 Z 0 1 -	BIT 2,D 2 8 Z 0 1 -	BIT 2,E 2 8 Z 0 1 -	BIT 2,H 2 8 Z 0 1 -	BIT 2,L 2 8 Z 0 1 -	BIT 2,(HL) 2 16 Z 0 1 -	BIT 2,A 2 8 Z 0 1 -	BIT 3,B 2 8 Z 0 1 -	BIT 3,C 2 8 Z 0 1 -	BIT 3,D 2 8 Z 0 1 -	BIT 3,E 2 8 Z 0 1 -	BIT 3,H 2 8 Z 0 1 -	BIT 3,L 2 8 Z 0 1 -	BIT 3,(HL) 2 16 Z 0 1 -	BIT 3,A 2 8 Z 0 1 -
6x	BIT 4,B 2 8 Z 0 1 -	BIT 4,C 2 8 Z 0 1 -	BIT 4,D 2 8 Z 0 1 -	BIT 4,E 2 8 Z 0 1 -	BIT 4,H 2 8 Z 0 1 -	BIT 4,L 2 8 Z 0 1 -	BIT 4,(HL) 2 16 Z 0 1 -	BIT 4,A 2 8 Z 0 1 -	BIT 5,B 2 8 Z 0 1 -	BIT 5,C 2 8 Z 0 1 -	BIT 5,D 2 8 Z 0 1 -	BIT 5,E 2 8 Z 0 1 -	BIT 5,H 2 8 Z 0 1 -	BIT 5,L 2 8 Z 0 1 -	BIT 5,(HL) 2 16 Z 0 1 -	BIT 5,A 2 8 Z 0 1 -
7x	BIT 6,B 2 8 Z 0 1 -	BIT 6,C 2 8 Z 0 1 -	BIT 6,D 2 8 Z 0 1 -	BIT 6,E 2 8 Z 0 1 -	BIT 6,H 2 8 Z 0 1 -	BIT 6,L 2 8 Z 0 1 -	BIT 6,(HL) 2 16 Z 0 1 -	BIT 6,A 2 8 Z 0 1 -	BIT 7,B 2 8 Z 0 1 -	BIT 7,C 2 8 Z 0 1 -	BIT 7,D 2 8 Z 0 1 -	BIT 7,E 2 8 Z 0 1 -	BIT 7,H 2 8 Z 0 1 -	BIT 7,L 2 8 Z 0 1 -	BIT 7,(HL) 2 16 Z 0 1 -	BIT 7,A 2 8 Z 0 1 -
8x	RES 0,B 2 8 - - - -	RES 0,C 2 8 - - - -	RES 0,D 2 8 - - - -	RES 0,E 2 8 - - - -	RES 0,H 2 8 - - - -	RES 0,L 2 8 - - - -	RES 0,(HL) 2 16 - - - -	RES 0,A 2 8 - - - -	RES 1,B 2 8 - - - -	RES 1,C 2 8 - - - -	RES 1,D 2 8 - - - -	RES 1,E 2 8 - - - -	RES 1,H 2 8 - - - -	RES 1,L 2 8 - - - -	RES 1,(HL) 2 16 - - - -	RES 1,A 2 8 - - - -
9x	RES 2,B 2 8 - - - -	RES 2,C 2 8 - - - -	RES 2,D 2 8 - - - -	RES 2,E 2 8 - - - -	RES 2,H 2 8 - - - -	RES 2,L 2 8 - - - -	RES 2,(HL) 2 16 - - - -	RES 2,A 2 8 - - - -	RES 3,B 2 8 - - - -	RES 3,C 2 8 - - - -	RES 3,D 2 8 - - - -	RES 3,E 2 8 - - - -	RES 3,H 2 8 - - - -	RES 3,L 2 8 - - - -	RES 3,(HL) 2 16 - - - -	RES 3,A 2 8 - - - -
Ax	RES 4,B 2 8 - - - -	RES 4,C 2 8 - - - -	RES 4,D 2 8 - - - -	RES 4,E 2 8 - - - -	RES 4,H 2 8 - - - -	RES 4,L 2 8 - - - -	RES 4,(HL) 2 16 - - - -	RES 4,A 2 8 - - - -	RES 5,B 2 8 - - - -	RES 5,C 2 8 - - - -	RES 5,D 2 8 - - - -	RES 5,E 2 8 - - - -	RES 5,H 2 8 - - - -	RES 5,L 2 8 - - - -	RES 5,(HL) 2 16 - - - -	RES 5,A 2 8 - - - -
Bx	RES 6,B 2 8 - - - -	RES 6,C 2 8 - - - -	RES 6,D 2 8 - - - -	RES 6,E 2 8 - - - -	RES 6,H 2 8 - - - -	RES 6,L 2 8 - - - -	RES 6,(HL) 2 16 - - - -	RES 6,A 2 8 - - - -	RES 7,B 2 8 - - - -	RES 7,C 2 8 - - - -	RES 7,D 2 8 - - - -	RES 7,E 2 8 - - - -	RES 7,H 2 8 - - - -	RES 7,L 2 8 - - - -	RES 7,(HL) 2 16 - - - -	RES 7,A 2 8 - - - -
Cx	SET 0,B 2 8 - - - -	SET 0,C 2 8 - - - -	SET 0,D 2 8 - - - -	SET 0,E 2 8 - - - -	SET 0,H 2 8 - - - -	SET 0,L 2 8 - - - -	SET 0,(HL) 2 16 - - - -	SET 0,A 2 8 - - - -	SET 1,B 2 8 - - - -	SET 1,C 2 8 - - - -	SET 1,D 2 8 - - - -	SET 1,E 2 8 - - - -	SET 1,H 2 8 - - - -	SET 1,L 2 8 - - - -	SET 1,(HL) 2 16 - - - -	SET 1,A 2 8 - - - -
Dx	SET 2,B 2 8 - - - -	SET 2,C 2 8 - - - -	SET 2,D 2 8 - - - -	SET 2,E 2 8 - - - -	SET 2,H 2 8 - - - -	SET 2,L 2 8 - - - -	SET 2,(HL) 2 16 - - - -	SET 2,A 2 8 - - - -	SET 3,B 2 8 - - - -	SET 3,C 2 8 - - - -	SET 3,D 2 8 - - - -	SET 3,E 2 8 - - - -	SET 3,H 2 8 - - - -	SET 3,L 2 8 - - - -	SET 3,(HL) 2 16 - - - -	SET 3,A 2 8 - - - -
Ex	SET 4,B 2 8 - - - -	SET 4,C 2 8 - - - -	SET 4,D 2 8 - - - -	SET 4,E 2 8 - - - -	SET 4,H 2 8 - - - -	SET 4,L 2 8 - - - -	SET 4,(HL) 2 16 - - - -	SET 4,A 2 8 - - - -	SET 5,B 2 8 - - - -	SET 5,C 2 8 - - - -	SET 5,D 2 8 - - - -	SET 5,E 2 8 - - - -	SET 5,H 2 8 - - - -	SET 5,L 2 8 - - - -	SET 5,(HL) 2 16 - - - -	SET 5,A 2 8 - - - -
Fx	SET 6,B 2 8 - - - -	SET 6,C 2 8 - - - -	SET 6,D 2 8 - - - -	SET 6,E 2 8 - - - -	SET 6,H 2 8 - - - -	SET 6,L 2 8 - - - -	SET 6,(HL) 2 16 - - - -	SET 6,A 2 8 - - - -	SET 7,B 2 8 - - - -	SET 7,C 2 8 - - - -	SET 7,D 2 8 - - - -	SET 7,E 2 8 - - - -	SET 7,H 2 8 - - - -	SET 7,L 2 8 - - - -	SET 7,(HL) 2 16 - - - -	SET 7,A 2 8 - - - -

Misc/control instructions

Jumps/calls

8bit load/store/move instructions

16bit load/store/move instructions

8bit arithmetic/logical instructions

16bit arithmetic/logical instructions

8bit rotations/shifts and bit instructions

Length in bytes →

INS reg
2 8
Z N H C

← Instruction mnemonic
← Duration in cycles
← Flags affected

Duration of conditional calls and returns is different when action is taken or not. This is indicated by two numbers separated by "/". The higher number (on the left side of "/") means duration of instruction when action is taken, the lower number (on the right side of "/") means duration of instruction when action is not taken.

Instruction **STOP** has according to manuals opcode **10 00** and thus is 2 bytes long. Anyhow it seems there is no reason for it so some assemblers code it simply as one byte instruction **10**.
Flags affected are always shown in **Z H N C** order. If flag is marked by "0" it means it is reset after the instruction. If it is marked by "1" it is set. If it is marked by "-" it is not changed. If it is marked by "Z", "N", "H" or "C" corresponding flag is affected as expected by its function.

d8 means immediate 8 bit data
d16 means immediate 16 bit data
a8 means 8 bit unsigned data, which are added to \$FF00 in certain instructions (replacement for missing **IN** and **OUT** instructions)
a16 means 16 bit address
r8 means 8 bit signed data, which are added to program counter

LD A,(C) has alternative mnemonic LD A,(\$FF00+C)
LD C,(A) has alternative mnemonic LD (\$FF00+C),A
LDH A,(a8) has alternative mnemonic LD A,(\$FF00+a8)
LDH (a8),A has alternative mnemonic LD (\$FF00+a8),A
LD A,(HL+) has alternative mnemonic LD A,(HLI) or LDI A,(HL)
LD (HL+),A has alternative mnemonic LD (HLI),A or LDI (HL),A
LD A,(HL-) has alternative mnemonic LD A,(HLD) or LDD A,(HL)
LD (HL-),A has alternative mnemonic LD (HLD),A or LDD (HL),A
LD HL,SP+r8 has alternative mnemonic LDHL SP,r8

https://www.pastraiser.com/cpu/gameboy/gameboy_opcodes.html

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Registers

15 ... 8	7 ... 0
A (accumulator)	F (flags)
B	C
D	E
H	L

15 ... 0
SP (stack pointer)
PC (program counter)

Flag register (F) bits:

7	6	5	4	3	2	1	0
Z	N	H	C	0	0	0	0

- Z** - Zero Flag
- N** - Subtract Flag
- H** - Half Carry Flag
- C** - Carry Flag
- 0** - Not used, always zero